

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	10
Total 1 mark			

Q2.

(a) (w=) $\frac{-6 \pm \sqrt{36 - 4(34)}}{2} \left\{ = \frac{-6 \pm \sqrt{-100}}{2} \right\}$

Correct substitution into quadratic formula OE

M1

$= \frac{-6 \pm 10i}{2}$

$\sqrt{-100} = 10i \text{ or } \sqrt{-100} / 2 = 5i$

B1

$= -3 \pm 5i$

$-3 \pm 5i \text{ (} p = -3, q = \pm 5 \text{)}$

NMS mark as 3/3 or 0/3

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(b) (i) $z = i(1 + i)(2 + i) = i(2 + 3i + i^2) = 2i + 3i^2 + i^3$

Attempt to expand all brackets.

M1

$= 2i + 3(-1) + i(-1)$

$i^2 = -1 \text{ used at least once}$

B1

$= -3 + i$

$-3 + i \text{ (} a = -3, b = 1 \text{)}$

A1

3

(ii) $z^* = -3 - i$

$-3 + i + m(-3 - i) = ni$

OE Ft c's a - bi

(A1)

$$\alpha = 4 - 5i$$

(A1)

(3)

(b) $\alpha + \beta + \gamma = -p$
 $-2 + 3i + 1 + 2i + 4 - 5i = 3$
 $(\Rightarrow p =) -3$

B1

1

(c) $\alpha\beta + \beta\gamma + \gamma\alpha = q$
 $q = \sum \alpha\beta$ and attempt to evaluate three products FT "their" α

$$(7 + 22i) + (-8 - i) + (14 + 3i) = q$$

M1

$$q = 13 + 24i$$

A1cao

2

[9]

Q5.

(a) (i) $x = \pm 3i$
 $\pm 3i$ ($a = 0, b = \pm 3$)

B1

1

(ii) $x = -2 \pm 3i$
If not correct, ft wrong answer(s) to (i) provided (i) has a non-zero b value

B1F

1

(b) (i) $(1 + x)^3 = 1 + 3x + 3x^2 + x^3$
Terms simplified in any order.

B1

(ii) $(1 + 2i)^3 = 1 + 3(2i) + 3(2i)^2 + (2i)^3 = 1 + 3(2i) + 3(4i^2) + (8i^3)$
Replacing x in (b)(i) by 2i, squaring and cubing correctly, only ft on c's wrong non-zero coefficients from (b)(i).

B1F

$$= 1 + 3(2i) + 3(4)(-1) + (8)(-i)$$

Use of $i^2 = -1$ at least once.

M1

$$= -11 - 2i$$

$$-11 - 2i \ (a = -11, b = -2)$$

A1

3

(iii) $z^* - z^3 = 1 - 2i - (-11 - 2i)$

$$\text{Use of } z^* = 1 - 2i$$

M1

$$= 12$$

If not correct, only ft on $1 - 2i - c$'s (b)(ii) if c 's (b)(ii) answer is of the form $a + bi$ with $a \neq 0$ and $b \neq 0$

A1F

2

[8]

Q6.

(a) Use of $(\sum \alpha)^2 = \sum \alpha^2 + 2\sum \alpha\beta$

AG

M1

A1

2

(b) $p = 0, q = 5 + 6i$

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B1,B1

2

(c) (i) Substitute $3i$ for z **or** use $3i\beta\gamma = -r$
allow for $3i\beta\gamma = r$

M1

$$-27i + 15i - 18 + r = 0 \text{ **or** } \beta\gamma = 5 + 6i + \alpha^2$$

any form

A1

$$r = 18 + 12i$$

one error

A1F

3

(ii) Cubic is $(z - 3i)(z^2 + 3iz - 4 + 6i)$

or use of $\beta\gamma$ and $\beta + \gamma$

clearly shown

M1A1

2

(iii) $f(-2) = 0$ or equate imaginary parts

M1

$$\beta = -2, \gamma = 2 - 3i$$

correct answers no working and no check B1 only

A1,A1F

3

[12]

Q7.

(a) $i(z + 7) + 3(z^* - i) = i(x + iy + 7) + 3(x - iy - i)$

M1 for use of $z^* = x - iy$

M1

$$= ix - y + 7i + 3x - 3iy - 3i$$

M1 for $i^2 y = -y$

M1

$$= 3x - y + i(x - 3y + 4)$$

If the five terms correct but not grouped into Real and Imaginary parts, allow A1 retrospectively provided the correct two expressions used in the M1 line in (b)

A1

3

(b) $3x - y = 0, \quad x - 3y + 4 = 0$

Attempting to equate all Real parts to zero and all Imaginary parts to zero

M1

$$x - 9x + 4 = 0 \quad (\text{or eg } y - 9y + 12 = 0)$$

A correct equation in either x or y
PI by correct final answer

A1

Solving to give $z = \frac{1}{2} + \frac{3}{2}i$

Allow $x = \frac{1}{2}, y = \frac{3}{2}$

A1

3

[6]

Q8.

(a) (i) $z_1^2 = \frac{1}{4} - i + i^2 = -\frac{3}{4} = i$
M1 for use of $i^2 = -1$

M1A1

2

(ii) $\text{LHS} = -\frac{3}{4} - i + \frac{1}{2} + i + \frac{1}{4} = 0$
AG; M1 for z^ correct*

M1A1

2

(b) $\text{LHS} = -\frac{3}{4} + i + \frac{1}{2} - i + \frac{1}{4} = 0$
AG; M1 for correct

M1A1

2

(c) $z \text{ real} \Rightarrow z^* = z$
Clearly stated

M1

Discr't zero or correct factorisation

AG

A1

2

[8]

Q9.

(a) $(1 + i)^2 = 2i$ or $(1 + i) = \sqrt{2}e^{\frac{\pi i}{4}}$

B1

$2i(1 + i) = 2i - 2$

AG

Alternative method:

$(1 + i)^3 = 1 + 3i^2 + 3i^3 + i^3$

B1

$= 2i - 2$

B1

B1

2

(b) (i) Substitute $z = 1 + i$

M1

Correct expansion

*allow for correctly picking out either the real
or the imaginary parts*

A1

$$k = -5$$

A1

3

(ii) $\beta + \gamma = 5 + i - \alpha = 4$
AG

B1

1

(iii) $\alpha\beta\gamma = 5(1 + i)$
allow if sign error

M1

$$\beta\gamma = 5$$

ft incorrect k

A1F

$$z^2 - 4z + 5 = 0$$

M1

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Use of formula or $(z - 2)^2 = -1$

No ft for real roots if error in k

A1F

$$z = 2 \pm i$$

A1F

5

NB allow marks for (b) in whatever order they appear

[11]

Q10.

(a) Use of $z^* = x - iy$

M1

$$(z - i)(z^* - i) = (x^2 + y^2 - 1) - 2ix$$

A1 may be earned in (b)

m1A1

3

(b) Equating R and I parts

M1

$$-2x = -8 \text{ so } x = 4$$

A1

$$16 + y^2 - 1 = 24 \text{ so } y = \pm 3 \text{ (} z = 4 \pm 3i \text{)}$$

A0 if $x = -4$ used

m1A1

4

[7]

Q11.

(a) $z^2 = 1 + 2i + i^2 = 2i$

M1 for use of $i^2 = -1$

M1A1

2

(b) $z^8 = (2i)^4$

or equivalent complete method

M1

$$\dots = 16i^4 = 16$$

convincingly shown (AG)

A1

2

(c) $(z^*)^2 = (1 - i)^2$

for use of $z^ = 1 - i$*

M1

$$\dots = -2i = -z^2$$

convincingly shown (AG)

A1

2

[6]

Q12.

(a) Use of $z^* = x - iy$

M1

Use of $i^2 = -1$

M1

$$(1 - 2i)z - z^* = 2y + i(2y - 2x)$$

A1 if one numerical error made

A2,1

4

(b) $2y = 20, 2y - 2x = 10$

equate and attempt to solve

M1

so $z = 5 + 10i$

allow $x = 5, y = 10$

A1

2

[6]

Q13.

(a) (i) $z^2 = (x^2 - 4) + i(4x)$

M1 for use of $i^2 = -1$

M1A1

R and I parts clearly indicated
*Condone inclusion of i in I part
 ft one numerical error*

A1F

3

(ii) $z^2 + 2z^* = (x^2 + 2x - 4) + i(4x - 4)$

*M1 for correct use of conjugate
 ft numerical error in (i)*

M1A1F

2

(b) $z^2 + 2z^*$ real if imaginary part zero

M1

... ie if $x = 1$

ft provided imaginary part linear

A1F

2

[7]

Q14.

$$z_1 + 4i z_1^* = (2 + i) + 4i(2 - i)$$

Use of conjugate

M1

$$\dots = (2 + i) + (8i + 4)$$

Use of $i^2 = -1$

M1

$$\dots = 6 + 9i, \text{ so } x = 6 \text{ and } y = 3$$

M1 for equating Real and imaginary parts

M1A1

[4]

Q15.

(a) (i) $\alpha + \beta = 2, \alpha\beta = 4$

B1B1

$$\alpha^3 + \beta^3 = (2)^3 - 3(4)(2) = -16$$

M1A1

$\alpha^3 \beta^3 = (4)^3 = 64$, hence result
convincingly shown (AG)

M1A1

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6

(ii) Discriminant 0, so roots equal
or by factorisation

B1E1

2

(b) $x = \frac{2 \pm \sqrt{4 - 16}}{2}$

or by completing square

M1

$$\dots = 1 \pm \frac{1}{2}i\sqrt{12}$$

A1

2

(c) $\alpha, \beta = 1 \pm i\sqrt{3}$
and $\alpha^3 = \beta^3$, hence result

E2

2

[12]

Q16.

- (a) Use of $z^* = x - iy$

M1

Use of $i^2 = -1$

M1

$$3iz + 2z^* = (2x - 3y) + i(3x - 2y)$$

Condone inclusion of i in I part

A1

3

- (b) Equating R and I parts

M1

$$2x - 3y = 7, 3x - 2y = 8$$

with attempt to solve

m1

$$z = 2 - i$$

Allow $x = 2, y = -1$

A1

3

[6]

Q17.

- (a) (i) $\alpha\beta\gamma = -18 + 12i$
accept $-(18 - 12i)$

B1

1

(ii) $\alpha + \beta + \gamma = 0$

B1

1

- (b) (i) $\alpha = -2$

B1F

1

(ii) $\beta\gamma = \frac{\alpha\beta\gamma}{\alpha} = 9 - 6i$

ft sign errors in (a) or (b)(i) or slips such as miscopy

M1A1F

2

(iii) $q = \sum \alpha\beta = \alpha(\beta + \gamma) + \beta\gamma$

M1

$= -2 \times 2 + 9 - 6i$

ft incorrect $\beta\gamma$ or α

$= 5 - 6i$

A1F

A1F

3

(c) $\beta = ki, \quad \gamma = 2 - ki$

B1

$ki(2 - ki) = 9 - 6i$

M1

$2k = -6 \quad (k^2 = 9) \quad k = -3$

imaginary parts

m1

$\beta = -3i, \quad \gamma = 2 + 3i$

A1

4

[12]

Q18.

(a) (i) Roots are $\pm 4i$

M1 for one correct root or two correct factors

M1A1

2

(ii) Roots are $1 \pm 4i$

M1 for correct method

M1A1

2

(b) (i) $(1+x)^3 = 1+3x+3x^2+x^3$
M1A0 if one small error

M1A1
2

(ii) $(1+i)^3 = 1+3i-3-i = -2+2i$
M1 if $i^2 = -1$ used

M1A1
2

(iii) $(1+i)^3 + 2(1+i) - 4i$
with attempt to evaluate

$\dots = (-2+2i) + (2-2i) = 0$
convincingly shown (AG)

M1
A1
2

[10]

Q19.

(a) $-k^3i + 2(1-i)(-k^2) + 32(1+i) = 0$
Any form

M1

Equate real and imaginary parts:

$-k^3 + 2k^2 + 32 = 0$

A1

$-2k^2 + 32 = 0$

A1

$k = \pm 4$

A1

$k = +4$

AG

E1

5

(b) Sum of roots is $-2(1-i)$

Or $\alpha\beta\gamma = -(32 + 32i)$
 Must be correct for M1

M1

Third root $2 - 2i$

A1✓

2

[7]

Q20.

(a) Use of $z^* = x - iy$

M1

$$z - 3iz^* = x + iy - 3ix - 3y$$

Condone sign error here

m1

$$R = x - 3y, I = -3x + y$$

Condone inclusion of i in I
Allow if correct in (b)

A1

3

(b) $x - 3y = 16, -3x + y = 0$

M1

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Elimination of x or y

m1

$$z = -2 - 6i$$

Accept $x = -2, y = -6$;
ft $x + 3y$ for $x - 3y$

A1F

3

[6]

Q21.

(a) (i) Full expansion of product

M1

Use of $i^2 = -1$

m1

$$(2 + \sqrt{5}i)(\sqrt{5} - i) = 3\sqrt{5} + 3i$$

$\sqrt{5}\sqrt{5} = 5$ must be used – Accept not fully simplified

A1

3

(ii) $z^* = x - iy (= \sqrt{5} + i)$

M1

Hence result

Convincingly shown (AG)

A1

2

(b) (i) Other root is $\sqrt{5} + i$

B1

1

(ii) Sum of roots is $2\sqrt{5}$

B1

Product is 6

M1A1

3

(iii) $p = -2\sqrt{5}, q = 6$

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B1

ft wrong answers in (ii)

B1ft

2

[11]

Q22.

(a) $\frac{1+i}{1-i} = \frac{(1+i)^2}{1-i^2} = i$

AG

M1A1

2

(b) $|z_2| = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1 = |z_1|$

M1A1

2

(c) $r = 1$

PI

B1

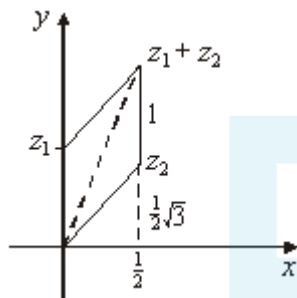
$$\theta = \frac{1}{2}\pi, \frac{1}{3}\pi$$

Deduct 1 mark if extra solutions

B1B1

3

(d)



Positions of the 3 points, relative to each other, must be approximately correct

B2,1F

2

(e) $\text{Arg}(z_1+z_2) = \frac{5}{12}\pi$

Clearly shown

B1

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$$\tan \frac{5}{12}\pi = \frac{1 + \frac{1}{2}\sqrt{3}}{\frac{1}{2}}$$

Allow if B0 earned

M1

$$= 2 + \sqrt{3}$$

AG must earn B0 for this

B1

3

[12]

Q23.

(a) $(z+i)^* = x-iy-i$

B2

2

(b) $\dots = 2ix - 2y + 1$
 $i^2 = -1$ used at some stage

M1

Equating R and I parts
 involving at least 5 terms in all

M1

$x = -2y + 1, -y - 1 = 2x$
 ft one sign error in (a)

A1ft

$z = -1 + i$
 ditto; allow $x = -1, y = 1$

m1A1ft

5

[7]

Q24.

(a) (i) $\alpha + \beta + \gamma = 4i$

B1

1

(ii) $\alpha\beta\gamma = 4 - 2i$

B1

1

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(b) (i) $\alpha + \alpha = 4i, \alpha = 2i$
 AG

B1

1

(ii) $\beta\gamma = \frac{4 - 2i}{2i} = -2i - 1$

Some method must be shown, eg $\frac{2}{i} - 1$

M1

AG

A1

2

(iii) $q = \alpha\beta + \beta\gamma + \gamma\alpha$

M1

$$= \alpha(\beta + \gamma) + \beta\gamma$$

Or $\alpha^2 + \beta\gamma$, ie suitable grouping

M1

$$= 2i \cdot 2i - 2i - 1 = -2i - 5$$

AG

A1

3

(c) Use of $\beta + \gamma = 2i$ and $\beta\gamma = -2i - 1$

M1

$$z^2 - 2iz - (1 + 2i) = 0$$

Elimination of say γ to arrive at
 $\beta^2 - 2i\beta - (1 + 2i) = 0$ *M1A0 unless*
also some reference to γ being a root
 AG

A1

2

(d) $f(-1) = 1 + 2i - 1 - 2i = 0$

For any correct method

M1

$$\beta = -1, \gamma = 1 + 2i$$

A1 for each answer

A1A1

3

[13]

Q25.

(a) $z^* = x - iy$

B1

1

(b) $R = 2x - y$

$$i^2 = -1 \text{ must be used}$$

B1

$$I = -x + 2y$$

Condone $I = i(x + 2y)$;

Answers may appear in (c)

B1
2

(c) Equating R and/or I parts

M1

Attempt to solve sim equations

m1

$$z = 1 + 2i$$

Allow $x = 1, y = 2$

A1
3

[6]

Q26.

(a) (i) $\alpha + \beta = 4, \alpha\beta = 13$

B1B1

(ii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

M1

$$\dots = 4^2 - 26 = -10$$

convincingly shown (AG)

A1
2

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(iii) The square of a real number is positive (or zero)

E1

The sum of two such squares is positive (or zero)

E1
2

(b) (i) $(\alpha + i) + (\beta + i) = 4 + 2i$
ft wrong value in (a)(i)

B1F
1

(ii) $(\alpha + i)(\beta + i) = 12 + 4i$
ditto

M1A1F
2

(c) Correct coeff of x or constant term

Using c's answers in (b)

M1

$$x^2 - (4 + 2i)x + (12 + 4i) = 0$$

ft wrong answers in (b)

A1F

2

[11]



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Examiner reports

Q1.

This question was answered very well, with almost all students selecting the correct answer of 10.

Q2.

Many correct answers were submitted for parts (a) and (b)(i) in this question which tested the complex numbers section of the specification. In part (b)(ii) a common mistake was an incorrect z^* value either $i + 3$ following $z = i - 3$ in (b)(i) or, more frequently, z^* written incorrectly as $i(1 - i)(2 - i)$. Students who obtained an incorrect value for z in part (b)(i) were awarded credit for a correct follow through value for z^* in part (b)(ii). Most students showed that they appreciated the need to equate the real parts and the imaginary parts for their resulting equation.

Q3.

This question which tested the complex numbers section of the specification proved, as expected, to be more demanding than the corresponding questions in recent MFP1 papers. In part (a)(i) a common wrong answer for $(z - 2i)^*$ was $x + iy + 2i$. Those students who started by writing $(x + iy - 2i)^*$ were generally more successful. There were some very good solutions to part (a)(ii), but some able students, having found the correct values for x and y , either stopped at that point or else gave their final answer as $\frac{1}{3} + \frac{4}{3}i$ which indicated that they thought that solving the given equation meant finding the value of $(z - 2i)^*$ and not the value of z . It was quite common to see some students making no attempt to equate real parts and imaginary parts, with final answers left in a form similar to $x + 4y - 3 + (2 - y - 4x)i$. The final part of the question was answered well by those who based their statement on:

- coefficients must be real for conjugate roots or
- sum of roots $(= 2p)$ cannot be $-10i$ if p is real or
- product of the roots $(p^2 + q^2)$ cannot be -29 if p and q are real.

However, correct statements were in the minority. Common wrong answers for example, 'it is not the difference of two squares' suggested that students were treating the roots $p \pm iq$ as the factors of the quadratic expression $z^2 + 10iz - 29$.

Q4.

This question was answered much better than similar questions in recent years, with very few making sign errors in either the sum or the product of the roots.

- (a) (i) Almost everyone wrote down the correct product of the roots as a complex number. Very few forgot the associated minus sign.
- (ii) The printed answer clearly helped with the product $(-2 + 3i)(1 + 2i)$ causing few problems. No doubt many did this calculation on a calculator and so in future questions involving algebra rather than simply numbers might be set.

- (iii) Some used simultaneous equations to find the real and imaginary parts of the complex number, α . Others wrote $\alpha = \frac{37-3i}{8+i}$ and many were able to write this in the correct form using a calculator. It had been expected that students would have shown their working when evaluating $\frac{37-3i}{8+i} \times \frac{8-i}{8-i}$. The correct answer was obtained by the majority of students.
- (b) Those who had the correct value of α were usually successful in finding the value of p . A small minority wrote $p = 3$, sometimes after obtaining $-p = 3$.
- (c) Some careless arithmetic was seen in this part but most students found the correct value of q . The majority found the sum of three separate products and a few realised that they could use the value of $\beta\gamma$ found earlier and evaluated an expression of the form $\beta\gamma + \alpha(\beta + \gamma)$, which was slightly more efficient.

Q5.

In this question on complex numbers, part (a) generally caused more problems than part (b).

In part (a)(i) a significant number of candidates failed to include the $\pm$, but, as follow through was accepted, most of these scored the mark in part (a)(ii).

Almost all candidates correctly expanded $(1 + x)^3$ in part (b)(i) and used their expansion within part (b)(ii) but there were many examples of incorrect / no squaring and cubing of the 2. The meaning and use of the complex conjugate z^* was well understood and a majority of candidates went on to give a correct follow through answer in part (b)(iii).

Q6.

This question proved to be the most popular and probably the question on which candidates could secure the highest marks. Parts (a) and (b) were well done, as was part (c)(i). If an error occurred in part(c)(i), it was usually either the cube of $3i$ written down as $-9i$ in the case where a candidate had substituted $3i$ for z in the cubic equation, or an error of sign in equating r to the product of the roots in the case when the roots of the cubic were used to find r . Solutions to part (c)(ii) were also pleasing, with many candidates showing that they needed to consider $\beta + \gamma$ as well as $\beta\gamma$ to show that β and γ were the roots of the quadratic equation. However a few thought that it was sufficient to show that one or other of $\beta + \gamma$ or $\beta\gamma$ satisfied the quadratic equation to establish the result. In part (c)(iii) many solutions were spoilt by errors of sign. For instance candidates who arrived at $\beta^2 = 4$, often just assumed that β was equal to 2.

Q7.

This question which tested complex numbers was generally answered very well with the majority of students scoring full marks. In part (a) almost all students correctly used $z^* = x - iy$ and $i^2 = -1$ and errors were generally restricted to either incorrect expansion of brackets or incorrect collecting of like terms. Part (b) caused slightly more problems with some students failing to even equate real parts and imaginary parts separately to zero.

Q8.

The first six marks in this question were obtained with apparent ease in the great majority of scripts. Part (c) usually produced no further marks as the candidates omitted the star from z^* without explaining why this was legitimate. Many candidates seemed to treat the given equation as a quadratic, despite the fact that two roots had already been found and two more were now being asked for.

Q9.

Again part (a) was well done. Many candidates scored well in part (b) too, obtaining the results to parts (i) and (ii) in a variety of ways, although sign errors, especially in the product of the roots, did lead to a loss of marks. One common error was to think that, because $1 + i$ was a root, $1 - i$ also had to be a root, in spite of the fact that the coefficients of the cubic were complex.

Q10.

Part (a) of this question was a minefield for some candidates, who showed the terms of their expansions very untidily and made errors, very often sign errors, in reaching a simplified form. In part (b), the technique of equating real and imaginary parts was well known and, in most cases, the two correct values were obtained for the imaginary part of z .

Q11.

Full marks were usually awarded in this question. Answers to part (b) were sometimes very laborious but eventually correct, but by contrast some answers were so brief as to be not totally convincing, earning one mark out of two. A few candidates fell short of full credit in part (c) by working on $(z^*)^2$ but not mentioning $-z^2$.

Q12.

The great majority of candidates showed that they had the necessary knowledge of complex numbers to cope with this very straightforward question. In a distressingly high number of instances the work was marred by elementary errors in the algebra, most commonly by a sign error causing $-z^*$ to appear as $-x - iy$. Many candidates also failed to indicate clearly in part (a) which were the real and imaginary parts, though many recovered the mark by using the real and imaginary parts correctly in part (b) of the question.

Q13.

In part (a) (i) of this question most candidates showed some knowledge of complex numbers but failed to display clearly the real and imaginary parts asked for in the question. Part (a) (ii) appeared to cause candidates no trouble at all; but by way of contrast, very few candidates saw what was required in part (b): many equated the real part to zero, while others equated the real and imaginary parts to each other.

Q14.

The great majority of candidates found this question a good starter and obtained full marks without too much trouble. Careless errors sometimes caused candidates to miss out on one of the method marks for the question.

Q15.

This question proved to be an excellent source of marks for the majority of candidates, apart from the discriminating test provided by part (c). Many candidates seemed to be familiar with the techniques relating to the cubes of the roots of a quadratic equation, though some struggled to find the correct expression for the sum of the cubes of the roots, while others lost marks by not showing enough evidence in view of the fact that the required equation was printed on the question paper.

Parts (a)(ii) and (b) proved straightforward for most candidates, but only a few were able to give a clear explanation in part (c), where many candidates claimed that the original roots α and β must be equal.

Q16.

Here again most candidates answered confidently and accurately. In part (a) many failed to state clearly which was the real part and which the imaginary part, though they recovered ground by using the correct expressions in part (b). As already stated, the solution of the simultaneous equations was often attempted by a substitution method.

Whichever method was used, numerical and sign errors were fairly common, but most candidates obtained the correct values for x and y . Two faults which were condoned this time were, in part (a), the retention of the factor 'i' in the imaginary part, and, in part (b) following correct values of x and y , a failure to give the final value of z correctly.

Q17.

There were many correct solutions to parts (a) and (b) of this question. Errors, if they did occur, were largely errors of sign. However, in part (c), a good number of candidates did not use the hint given in the question to replace β by ki but instead called the two remaining roots β and $2 - \beta$. A quadratic equation in β ensued followed by the use of the quadratic formula and ended up with the square root of a complex number at which point solutions were abandoned.

Q18.

This opening question gave almost all the candidates the opportunity to score a high number of marks. Even when careless errors were made, for example the omission of the plus-or-minus symbol in one or both sections of part (a), there was much correct work for the examiners to reward. The expansion of the cube of a binomial expression in part (b)(i) seemed to be tackled more confidently than in the past. Almost all the candidates used $i^2 = -1$ in part (b)(ii), though i^3 some were unsure how to deal with.

Q19.

Although this question was attempted by almost every candidate, there were few whose solutions presented the rigour required. Most substituted ki for z in the cubic equation, equated real parts and subsequently wrote $z^2 = 16$ so $z = 4$, not realising that $z = -4$ was also a root of $z^2 = 16$ and that imaginary parts had to be equated in order to reject the solution $z = -4$. Part (b) was not particularly well answered either. The most common errors were errors of sign in the use of $\alpha + \beta + \gamma$ or $\alpha\beta\gamma$, and those candidates using the product of the roots made extra work for themselves as they obtained a rational expression for γ which needed to be simplified. Some candidates thought that γ equalled $-4i$, the complex conjugate of α .

Q20.

Although most of the candidates who took this paper are good at algebra, there are still quite a number who made elementary sign errors. The fourth term of the expansion in part (a) frequently came out with a plus instead of a minus. The error of including an 'i' in the imaginary part was condoned.

In part (b) some candidates seemed not to realise that they needed to equate real and imaginary parts, despite the hint in part (a). Others used 16 for both the real and imaginary parts of the right-hand side of the equation. But a large number solved correctly to obtain full marks. Those who had made the sign error already referred to were given full marks in part (b) if their work was otherwise faultless.

Q21.

Part (a)(i) was answered well by the majority of candidates, though the fourth term in the expansion often appeared as $-i^2$ instead of $-i^2 \sqrt{5}$, and $-i^2$ was frequently simplified to -1 in the next line.

The next part, (a)(ii), tested the candidates' knowledge of technique to some extent. Those who understood what they were being asked to do invariably realised that the necessary hard work had already been done in part (a)(i), so that very little writing was needed to pick up two further marks. Those who thought they had to prove, rather than verify, the given statement were awarded one mark for showing a knowledge of the conjugate of z , but then spent much valuable time trying to solve the equation. There were three indications that this was not what the question required: the word 'Hence', the word 'verify', and the award of only two marks.

Part (b) of this question was generally well answered, though some candidates did not seem to be aware of the implications of the statement that the coefficients in the quadratic equation were real. The answers to part (b)(ii) were sometimes given in terms of p and q , and in part (b)(iii) many candidates equated p to the sum of the roots instead of to this number preceded by a minus sign.

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Q22.

Part (a) was well done, as was part (b). In part (c) it was surprising to note how many candidates could not express the complex number in the form $re^{i\theta}$, although z_2 was almost

invariably correctly written as $e^{i\frac{\pi}{3}}$. Errors in part (c) however did not deter candidates from drawing a correct Argand diagram as they usually used the form $a + ib$ when plotting their

points. Although the vast majority of scripts ended with $\tan \frac{5\pi}{12} = 2 + \sqrt{3}$, very, very few

candidates gave convincing proof that $\arg(z_1 + z_2)$ was $\frac{5\pi}{12}$, but rather seemed to take it for granted.

Q23.

Part (a) of this question seemed to take many candidates unawares. They knew that the conjugate of z was $x - iy$, but claimed that the conjugate of $z + i$ was $x - iy + i$ or that it was simply $z - i$. Most candidates were aware of the required approach to part (b). They expanded the right-hand side of the equation and then equated the real and imaginary parts, giving two simple simultaneous equations which they could solve. In many cases their efforts were marred by sign errors or by the omission of a 2 when expanding the term

2iz.

Q24.

Apart from the occasional sign errors, part (a) was answered well. Where sign errors did occur there was some faking to establish the printed answers in part (b). Part (b) is an example of what was mentioned at the beginning of this report in that with all three answers being printed, sufficient working needed to be shown in order to obtain full credit. Whilst most candidates knew roughly what was required for part (c), few candidates could express their argument succinctly. A number of candidates attempted to divide the cubic equation by $z-2i$ with varying success. Probably the commonest method of approach in part (d) was to substitute β for z in the quadratic equation in z and then to equate real parts. Equating real parts led to $\beta^2 = 1$ from which a substantial number of candidates assumed that $\beta = 1$ instead of considering the imaginary parts of the equation as well.

Q25.

Many candidates showed a sufficient familiarity with complex numbers to pick up a fair number of marks in this question. Part (a) was almost always correct, but a sign error spoiled many answers to part (b). Many candidates did not clearly state the real and imaginary parts in part (b), or stated the imaginary part as the correct answer multiplied by i , but these faults were condoned if, as usually happened, the candidate used the real and imaginary parts correctly in part (c).

Q26.

There were many high-scoring attempts at this question, most candidates showing the necessary familiarity with the roots and coefficients of quadratic equations as well as with complex numbers. The explanation called for in part (a)(iii) often showed an awareness that real numbers cannot have negative squares, but failed to indicate the fact that $\alpha + \beta^2$ was not simply the square of α or of β . Some reference to the addition of the two squares was needed for full credit here.

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In part (b) some candidates gave answers involving α and β rather than using the results they had obtained in part (a). In part (c) many candidates lost a mark by giving the wrong sign for the x term and/or omitting the 'equals zero' from their 'equation'.